

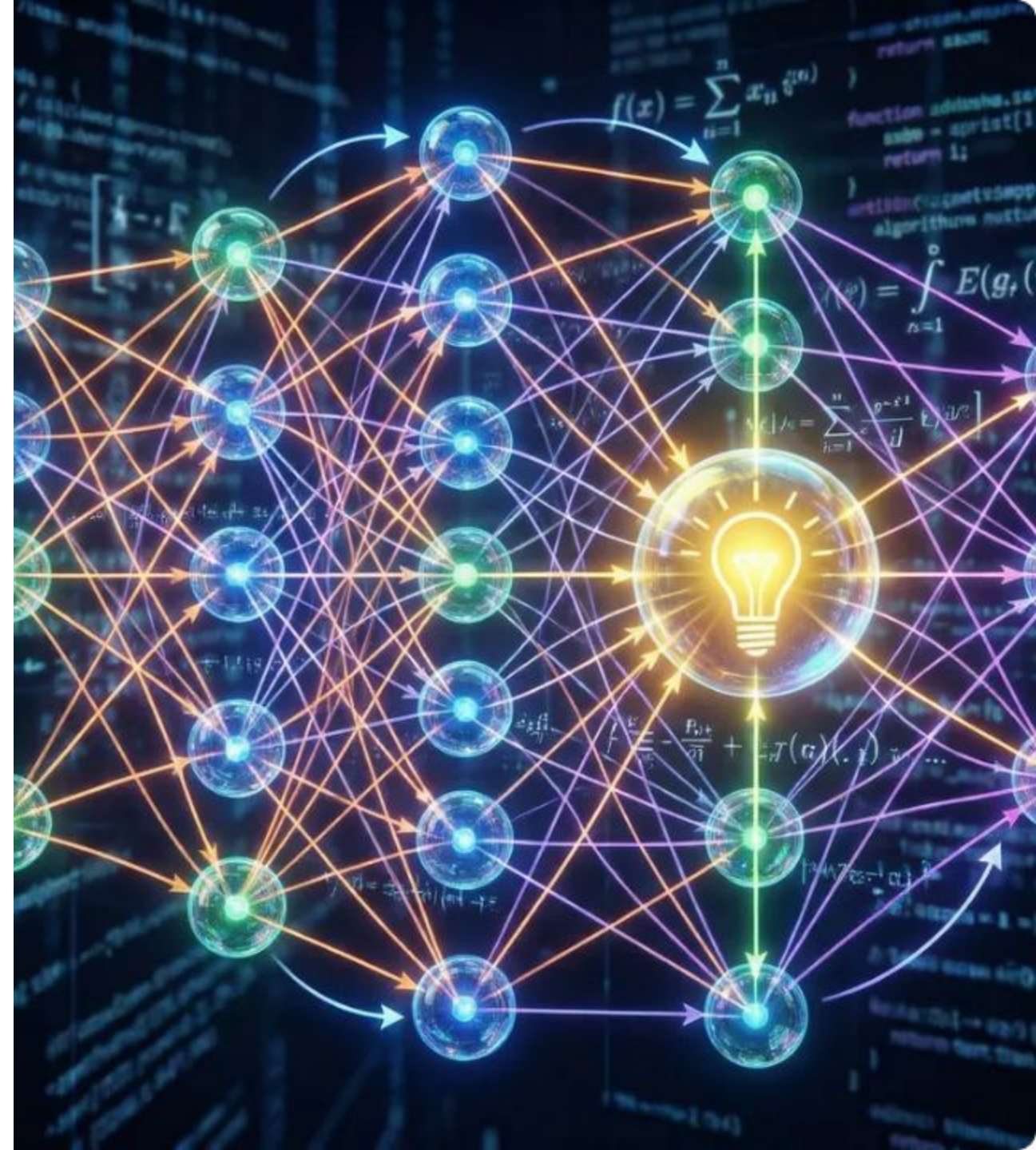
Modeling Credit Risk

Developing a Macro-Conditioned Probability of Default
Transition Matrix Model

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Project Overview

- Credit rating migrations forecasted using Markov Chains
- 1-quarter (3-month) time horizon for high-resolution tracking
- Quantifies likelihood of rating migrations and default



Methodology & Tech Stack

Built with Python & Macro Data



Python Ecosystem

- End-to-end model development entirely in Python
- **pandas** utilized for data wrangling and matrix manipulation
- **statsmodels** applied for econometric equations
- **scikit-learn** leveraged for unsupervised learning tasks



Industry Segmentation

- Borrowers segmented by industry to build targeted models
- **K-means clustering** applied to group similar industry risk profiles
- Captures distinct sector-specific sensitivities to economic cycles



Macro-Conditioning

- Segment models conditioned on broader macroeconomic variables
- Data-driven variable selection via rigorous statistical testing
- Incorporated checks for stationarity and predictive power

The 1-Quarter Transition Matrix

Macro-Conditioned Output Data (%)

Initial \ Final	AAA	AA	A	BBB	BB	B	CCC	Default
AAA	98.50	1.30	0.15	0.03	0.02	0.00	0.00	0.00
AA	0.20	97.80	1.70	0.20	0.05	0.03	0.02	0.00
A	0.05	0.60	97.50	1.50	0.20	0.10	0.03	0.02
BBB	0.01	0.10	2.50	94.00	2.80	0.50	0.05	0.04
BB	0.01	0.05	0.30	2.50	92.50	4.00	0.40	0.24
B	0.00	0.02	0.10	0.30	3.50	93.00	2.00	1.08
CCC	0.00	0.00	0.05	0.20	1.00	4.50	88.00	6.25

Core Mathematical Properties

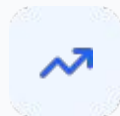
Markov Assumptions



The Markov Property

- Future states depend solely on current state
- Historical paths are irrelevant (memoryless property)

$$P(X_{t+1} = j | X_t = i) = p_{ij}$$



Diagonal Dominance

- Extremely high probabilities on matrix diagonal
- Reflects massive rating 'stickiness' over short horizon
- Majority of entities maintain stable ratings



Absorbing States

- 'Default' treated as an absorbing state
- Entities entering default remain permanently
- Probability of transitioning out of default is zero

Visualizing Migration Risk

Starting State: 'BBB' Rating

Forecasted probability distribution for a 'BBB' rated borrower after 1 quarter:



Image Sources



https://media.easy-peasy.ai/7cd52b37-a38b-4e6c-9ae7-91d3588a7d76/23d4d454-ceda-4490-aaeb-ae86256eed64_medium.webp

Source: easy-peasy.ai